

HAFSA ANWAR

Robotics & Intelligence Systems Student | Software Development • Automation • Computer Vision

Karachi, Pakistan | hafsaa.anwar017@gmail.com | 0330-1396322 | [GitHub](#) | [LinkedIn](#)

PROFILE

Robotics & Intelligence Systems undergraduate with hands-on software engineering experience building REST APIs, automation workflows, and computer vision systems in Python. Comfortable across the stack from backend services (FastAPI, Flask) to workflow automation (n8n) and lightweight web front-ends (JavaScript, HTML, CSS) with additional experience in robotics simulation using ROS 2 and C++. Quick to pick up new tools, ships working systems, and enjoys turning ambiguous problems into clean, testable code.

TECHNICAL SKILLS

AI/ML & Data: Machine Learning, Deep Learning, CNNs, Feature Engineering, Model Development, Generative AI, Prompt Engineering, RAG, LLM Applications, OpenCV, YOLO, MediaPipe, Object Detection & Tracking, Image Processing, OCR, Data Analysis

Robotics: ROS 2, MoveIt, RViz, Gazebo, Forward/Inverse Kinematics, Trajectory Planning, A* Path Planning, LiDAR Navigation, MATLAB, SolidWorks, AutoCAD, Xilinx, Circuit Lab, NI Multisim

Tools & Frameworks: Python, C++, C, JavaScript, HTML, CSS, FastAPI, Flask, REST APIs, API/Swagger, MLflow, Git, GitHub, VS Code, Linux

Automation: n8n, Make, Playwright, Automation Anywhere, Workflow Automation, MongoDB Atlas Vector Search, ChromaDB

EXPERIENCE

AI & Automation Engineer - Tandorge Limited

June 2026 - Present

- Building a Facebook Lead Extractor / Lead Intelligence system that automates extraction of business leads from Facebook using browser automation (Python, FastAPI, Playwright, ChromaDB) - in progress.
- Building a custom ERP system (Python backend, PostgreSQL) covering employee/HR management, production tracking, wastage tracking, and reporting, rolled out in phases.

AI Automation Intern - DevelopersHub Corporation

August 2026 - Present | Remote

- Built an AI-powered support ticket triage system that classifies incoming requests, validates LLM outputs, and scores confidence/risk before routing uncertain cases to human review.
- Learned and applied Playwright for browser automation.

Gen AI Fellow - Mind Works International

July 2026 - August 2026 | Remote

- Built workflow automation pipelines using n8n, connecting language-model-based systems with external APIs and data sources.
- Applied retrieval-augmented generation techniques within automation workflows to structure and route information.

Robotics & Automation Intern - Decode Labs

June 2026 - July 2026 | Remote

- 6-DOF Robotic Arm Simulation: designed a simulation using Forward & Inverse Kinematics with trajectory planning and obstacle avoidance, integrated with ROS 2 tools (MoveIt, RViz, Gazebo).
- Gear Inspection System: built a computer vision pipeline in OpenCV to detect gear defects on a conveyor belt, classifying parts as PASS/FAIL using contour analysis.
- Autonomous Mobile Robot: developed a LiDAR-based mapping and navigation system using A* path planning with dynamic replanning for obstacle avoidance in simulated maze environments.

PROJECTS

Support Ticket Triage System

Python • Workflow Automation • Validation • Human-in-the-Loop

- Built an automated support ticket pipeline that classifies incoming requests using rule-based and model-driven logic.
- Added validation and confidence/risk scoring on model outputs, routing uncertain cases to human review instead of auto-resolving them.

HerSafeAI : Harassment Reporting Assistant

n8n • JavaScript • HTML • CSS - live: [hersafeai.netlify.app](#)

- Built a conversational assistant that guides users through reporting harassment incidents step by step.
- Implemented risk/confidence scoring to route edge cases to human review, validating accuracy across 5+ ticket scenarios..

Facebook Lead Extractor

Python • FastAPI • Playwright • ChromaDB — In Progress

- Building an AI-powered lead intelligence system that automates extraction of business leads from Facebook using browser automation.
- Structuring extracted data and storing it in a vector database (ChromaDB) for fast retrieval and downstream automation workflows.

GitHub Commit Notifier

Python • Flask • GitHub Webhooks • Cloudflare Tunnel — [github.com/Hafsa-Anwar65/github-commit-notifier](#)

- Built a Flask server that listens for GitHub webhook events (pushes, pull requests) and prints real-time formatted notifications to the terminal.
- Exposed the local server publicly using a Cloudflare Tunnel, with no cloud hosting required.

CERTIFICATIONS

AI Fundamentals (IBM) | Python (UniAthena) | Machine Learning (MathWorks) | Prompt Engineering (SoloLearn) | Generative AI (LinkedIn Learning) | Data Visualization (Kaggle) | Building RAG Apps Using MongoDB (MongoDB) | Data Science & Analytics (HP LIFE) | Introduction to Developing AI Agents (Microsoft)

EDUCATION

BS in Robotics & Intelligence Systems - Bahria University

Sept 2024 - Present (5th Semester) | Karachi

Higher Secondary Education - Bahria College Karsaz

2022 - 2024 | Karachi

Languages: English, Urdu, Italian